Chapter 1

Nonparametric regression: A Summary

Fully Bayesian Regression Recap

- In fully Bayesian regression, no specific model parameters $\overline{\mathbf{w}}^*$ are identified. Instead, the predictive distribution $p(t|\mathbf{x})$ is derived.
- Under Gaussian assumptions:

$$p(t|\mathbf{x}, \mathbf{X}, \mathbf{t}, \alpha, \beta) = \mathcal{N}(t|m(\mathbf{x}), \sigma^2(\mathbf{x})) \tag{1.1}$$

with:

$$m(\mathbf{x}) = \beta \overline{\mathbf{x}}^T \mathbf{S} \overline{\mathbf{X}}^T \mathbf{t}$$
$$\sigma^2(\mathbf{x}) = \frac{1}{\beta} + \overline{\mathbf{x}}^T \mathbf{S} \overline{\mathbf{x}}$$

where $\mathbf{S} = (\alpha \mathbf{I} + \beta \overline{\mathbf{X}}^T \overline{\mathbf{X}})^{-1} \in \mathbb{R}^{(d+1)\times(d+1)}$.

Equivalent Kernel

- Prediction (expectation of predictive distribution):

$$h(\mathbf{x}) = m(\mathbf{x}) = \beta \overline{\mathbf{x}}^T \mathbf{S} \overline{\mathbf{X}}^T \mathbf{t} \tag{1.2}$$

- Since $\overline{\mathbf{X}}^T \mathbf{t} = \sum_{i=1}^n \overline{\mathbf{x}}_i t_i$, we can write:

$$h(\mathbf{x}) = \sum_{i=1}^n \beta \overline{\mathbf{x}}^T \mathbf{S} \overline{\mathbf{x}}_i t_i \tag{1.3}$$

- Prediction is a linear combination of target values t_i, with weights depending on $\mathbf{x}$ and $\mathbf{x}_i$.
- Define equivalent kernel $\kappa(\mathbf{x}_1, \mathbf{x}_2) = \beta \overline{\mathbf{x}}_1^T \mathbf{S} \overline{\mathbf{x}}_2$.
- Then:

$$h(\mathbf{x}) = \sum_{i=1}^n \kappa(\mathbf{x}, \mathbf{x}_i) t_i \tag{1.4}$$

- With basis functions $\boldsymbol{\phi}$:

$$\kappa(\mathbf{x}_1, \mathbf{x}_2) = \beta \boldsymbol{\phi}(\mathbf{x}_1)^T \mathbf{S} \boldsymbol{\phi}(\mathbf{x}_2) \tag{1.5}$$

- $\kappa(\mathbf{x}, \mathbf{x}_i)$ measures relevance of known target t_i for predicting at $\mathbf{x}$.
- Equivalent kernel tends to assign greater relevance to targets t_i corresponding to items $\mathbf{x}_i$ near $\mathbf{x}$ (localization property).
- Instead of introducing basis functions, we can directly use a **localized kernel** defined on $\mathbb{R}^d \times \mathbb{R}^d$.

Kernel Regression

- Predict target for $\mathbf{x}$ by referring to training items, especially those closer to $\mathbf{x}$.
- Controlled by predefined **kernel** function $\kappa_h(\mathbf{x})$ (non-negligible only near 0).
- Common kernel: Gaussian (RBF) kernel:

$$g(\mathbf{x}) = e^{-\frac{\|\mathbf{x}\|^2}{2h^2}} \tag{1.6}$$

- Goal: approximate conditional expectation:

$$\mathbb{E}\left[t|\mathbf{x}\right] = \frac{\int t \; p(\mathbf{x}, t) dt}{\int p(\mathbf{x}, t) dt} \tag{1.7}$$

- Approximate joint distribution using kernel:

$$p(\mathbf{x}, t) \approx \frac{1}{n} \sum_{i=1}^{n} \kappa_h(\mathbf{x} - \mathbf{x}_i) \kappa_h(t - t_i) \tag{1.8}$$

- Assuming kernel is non-negative, has mean 0 and area 1 (probability density), we get **Nadaraya-Watson** estimator:

$$h(\mathbf{x}) = \frac{\sum_{i=1}^{n} \kappa_h(\mathbf{x} - \mathbf{x}_i) t_i}{\sum_{i=1}^{n} \kappa_h(\mathbf{x} - \mathbf{x}_i)} = \sum_{i=1}^{n} w_i(\mathbf{x}) t_i \tag{1.9}$$

  with $w_i(\mathbf{x}) = \frac{\kappa_h(\mathbf{x} - \mathbf{x}_i)}{\sum_{j=1}^{n} \kappa_h(\mathbf{x} - \mathbf{x}_j)}$.
- With basis functions $\boldsymbol{\phi}$:

$$h(\mathbf{x}) = \sum_{i=1}^{n} w_i(\boldsymbol{\phi}(\mathbf{x})) t_i, \quad w_i(\mathbf{x}) = \frac{\kappa_h(\boldsymbol{\phi}(\mathbf{x}) - \boldsymbol{\phi}(\mathbf{x}_i))}{\sum_{j=1}^{n} \kappa_h(\boldsymbol{\phi}(\mathbf{x}) - \boldsymbol{\phi}(\mathbf{x}_i))} \tag{1.10}$$

Locally Weighted Regression (LOESS)

- Improves Nadaraya-Watson by using weighted version of sum of squared differences loss.

- For predicting t at $\mathbf{x}$, define local loss with weights $\psi_i(\mathbf{x}) = \kappa_h(\mathbf{x} - \mathbf{x}_i)$:

$$L(\mathbf{x}) = \sum_{i=1}^{n} \psi_i(\mathbf{x})(\overline{\mathbf{w}}^T \overline{\mathbf{x}}_i - t_i)^2 \tag{1.11}$$

- Minimization yields:

$$\overline{\mathbf{w}}^*(\mathbf{x}) = (\overline{\mathbf{X}}^T \Psi(\mathbf{x}) \overline{\mathbf{X}})^{-1} \overline{\mathbf{X}}^T \Psi(\mathbf{x}) \mathbf{t} \tag{1.12}$$

  where $\Psi(\mathbf{x})$ is diagonal with $\Psi(\mathbf{x})_{ii} = \psi_i(\mathbf{x})$.
- Prediction:

$$h(\mathbf{x}) = \overline{\mathbf{w}}^*(\mathbf{x})^T \overline{\mathbf{x}} \tag{1.13}$$

Local Logistic Regression

- Same approach applied to classification using weighted cross-entropy:

$$L(\mathbf{x}) = \sum_{i=1}^{n} \kappa_h(\mathbf{x} - \mathbf{x}_i)(t_i \log p_i - (1 - t_i) \log(1 - p_i)) \tag{1.14}$$

  with $p_i = \sigma(\overline{\mathbf{w}}^T \overline{\mathbf{x}}_i)$.

Gaussian Processes

Definition

- A **stochastic process** $f(\mathbf{x})$ is a collection of random variables $\{f(\mathbf{x}) : \mathbf{x} \in \chi\}$.
- It is a **Gaussian process** (GP) if for any finite subset $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$, the values $f(\mathbf{x}_1), \dots, f(\mathbf{x}_n)$ have a joint multivariate Gaussian distribution.
- GP generalizes multivariate Gaussians to infinite dimensions.
- Specified by:
  - **Mean function** $m(\mathbf{x})$: for any $\mathbf{x}$, $\mathbb{E}[f(\mathbf{x})] = m(\mathbf{x})$.
  - **Covariance function** (kernel) $\kappa(\mathbf{x}, \mathbf{x}')$: $\mathrm{Cov}(f(\mathbf{x}), f(\mathbf{x}')) = \kappa(\mathbf{x}, \mathbf{x}')$.
- For any finite set $\mathbf{X}$, $f(\mathbf{X}) \sim \mathcal{N}(\boldsymbol{\mu}_\mathbf{X}, \boldsymbol{\Sigma}_\mathbf{X})$ where:
  - $\boldsymbol{\mu}_\mathbf{X} = (m(\mathbf{x}_1), \dots, m(\mathbf{x}_n))^T$
  - $\boldsymbol{\Sigma}_\mathbf{X}[i, j] = \kappa(\mathbf{x}_i, \mathbf{x}_j)$ (Gram matrix $G_\mathbf{X}$)
- κ must be a **positive definite kernel**: for any distinct $\mathbf{x}_1, \dots, \mathbf{x}_n$ and constants c_i not all zero:

$$\sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j \kappa(\mathbf{x}_i, \mathbf{x}_j) > 0 \tag{1.15}$$

- Common kernel: RBF kernel $\kappa(\mathbf{x}_1, \mathbf{x}_2) = \sigma^2 e^{-\frac{\|\mathbf{x}_1 - \mathbf{x}_2\|^2}{2\tau^2}}$.
- RBF kernel assigns higher covariance to nearby points, producing smooth functions. Smoothing increases with τ.

Posterior Distribution (Noise-Free Case)

- GP defines a prior distribution over functions: $f \sim \mathcal{GP}(m, \kappa)$.
- Assume we observe $t_i = f(\mathbf{x}_i)$ exactly (no noise) at training points $\mathbf{X}$.
- For new points $\mathbf{Z}$, joint distribution of $f(\mathbf{X})$ and $f(\mathbf{Z})$ is:

$$\begin{pmatrix} f(\mathbf{X}) \\ f(\mathbf{Z}) \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \boldsymbol{\mu}_\mathrm{X} \\ \boldsymbol{\mu}_\mathrm{Z} \end{pmatrix}, \begin{pmatrix} G_\mathrm{X} & G_{\mathrm{Z,X}} \\ G_{\mathrm{Z,X}}^T & G_\mathrm{Z} \end{pmatrix} \right)$$

  where $G_{\mathrm{Z,X}}[i, j] = \kappa(\mathbf{z}_i, \mathbf{x}_j)$.
- Using properties of conditional Gaussians, predictive distribution $p(f(\mathbf{Z})|\mathbf{Z}, \mathbf{X}, f(\mathbf{X}))$ is Gaussian with:

$$\begin{aligned} \boldsymbol{\mu}_{pr} &= \boldsymbol{\mu}_\mathrm{Z} + G_{\mathrm{Z,X}} G_\mathrm{X}^{-1} (f(\mathbf{X}) - \boldsymbol{\mu}_\mathrm{X}) \\ \boldsymbol{\Sigma}_{pr} &= G_\mathrm{Z} - G_{\mathrm{Z,X}} G_\mathrm{X}^{-1} G_{\mathrm{Z,X}}^T \end{aligned}$$

- For a single test point $\mathbf{z}$:

$$\begin{aligned} \mu_{pr} &= m(\mathbf{z}) + G_{\mathbf{z},\mathrm{X}} G_\mathrm{X}^{-1} (\mathbf{t} - \boldsymbol{\mu}_\mathrm{X}) \\ \sigma_{pr}^2 &= \kappa(\mathbf{z}, \mathbf{z}) - G_{\mathbf{z},\mathrm{X}} G_\mathrm{X}^{-1} G_{\mathbf{z},\mathrm{X}}^T \end{aligned}$$

- With zero mean assumption ($m(\mathbf{x}) = 0$):

$$\mu_{pr} = G_{\mathbf{z},\mathrm{X}} G_\mathrm{X}^{-1} \mathbf{t} \tag{1.16}$$

- In noise-free case, predictive mean interpolates training points exactly: for $\mathbf{x}_i \in \mathbf{X}$, $\mu_{pr}(\mathbf{x}_i) = t_i$ and $\sigma_{pr}^2(\mathbf{x}_i) = 0$.

Gaussian Process Regression with Gaussian Noise

- Realistic assumption: $t_i = f(\mathbf{x}_i) + \varepsilon$ with independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, \sigma_f^2)$.
- Then $\mathrm{Cov}(t_i, t_j) = \kappa(\mathbf{x}_i, \mathbf{x}_j) + \sigma_f^2 \delta_{ij}$.
- Covariance matrix for training points becomes:

$$\hat{\boldsymbol{\Sigma}}_\mathrm{X} = G_\mathrm{X} + \sigma_f^2 \mathbf{I} \tag{1.17}$$

- For new points $\mathbf{Z}$, joint distribution:

$$\begin{pmatrix} \mathbf{t} \\ f(\mathbf{Z}) \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \boldsymbol{\mu}_\mathrm{X} \\ \boldsymbol{\mu}_\mathrm{Z} \end{pmatrix}, \begin{pmatrix} \hat{\boldsymbol{\Sigma}}_\mathrm{X} & G_{\mathrm{Z,X}} \\ G_{\mathrm{Z,X}}^T & G_\mathrm{Z} \end{pmatrix} \right)$$

- Predictive distribution $p(f(\mathbf{Z})|\mathbf{Z}, \mathbf{X}, \mathbf{t})$ is Gaussian with:

$$\begin{aligned} \hat{\boldsymbol{\mu}}_{pr} &= \boldsymbol{\mu}_\mathrm{Z} + G_{\mathrm{Z,X}} \hat{\boldsymbol{\Sigma}}_\mathrm{X}^{-1} (\mathbf{t} - \boldsymbol{\mu}_\mathrm{X}) \\ \hat{\boldsymbol{\Sigma}}_{pr} &= G_\mathrm{Z} - G_{\mathrm{Z,X}} \hat{\boldsymbol{\Sigma}}_\mathrm{X}^{-1} G_{\mathrm{Z,X}}^T \end{aligned}$$

- With zero mean assumption:

$$\hat{\boldsymbol{\mu}}_{pr} = G_{\mathrm{Z,X}} \hat{\boldsymbol{\Sigma}}_\mathrm{X}^{-1} \mathbf{t} \tag{1.18}$$

- For a single test point $\mathbf{z}$:

$$\begin{aligned} \mu_{pr} &= m(\mathbf{z}) + G_{\mathrm{z,X}} \hat{\boldsymbol{\Sigma}}_\mathrm{X}^{-1} (\mathbf{t} - \boldsymbol{\mu}_\mathrm{X}) \\ \sigma_{pr}^2 &= \kappa(\mathbf{z}, \mathbf{z}) - G_{\mathrm{z,X}} \hat{\boldsymbol{\Sigma}}_\mathrm{X}^{-1} G_{\mathrm{z,X}}^T \end{aligned}$$

- With noise, predictive mean no longer interpolates exactly, and predictive variance at training points is $\sigma_f^2 > 0$.

Estimating Kernel Parameters

- Predictive performance depends on suitability of chosen kernel and its hyperparameters.
- For RBF kernel:

$$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \sigma_f^2 e^{-\frac{1}{2}(\mathbf{x}_i - \mathbf{x}_j)^T \mathbf{M} (\mathbf{x}_i - \mathbf{x}_j)} + \sigma_y^2 \delta_{ij} \tag{1.19}$$

- Simplest $\mathbf{M} = l^{-2}\mathbf{I}$, giving hyperparameters σ_f, σ_y, l.
- Varying these parameters significantly affects predictions:
  - l (length scale): controls smoothness (larger $l \Rightarrow$ smoother).
  - σ_f (signal variance): controls amplitude of function variations.
  - σ_y (noise variance): controls amount of assumed noise.
- Hyperparameters can be estimated by maximizing the log marginal likelihood:

$$\log p(\mathbf{t}|\mathbf{X}) = -\frac{1}{2} \mathbf{t}^T \hat{\boldsymbol{\Sigma}}_\mathrm{X}^{-1} \mathbf{t} - \frac{1}{2} \log |\hat{\boldsymbol{\Sigma}}_\mathrm{X}| - \frac{n}{2} \log 2\pi \tag{1.20}$$

- This provides an automatic trade-off between data fit and model complexity.

Recap of Gaussian Process Properties

- GP provides a nonparametric Bayesian approach to regression.
- Directly defines a distribution over functions via mean and covariance functions.
- Predictions come with natural uncertainty estimates (predictive variance).
- Kernel choice encodes assumptions about function smoothness and structure.
- Hyperparameters can be learned from data via marginal likelihood optimization.
- GP regression is equivalent to Bayesian linear regression with an infinite number of basis functions.
- The equivalent kernel perspective shows GP predictions are weighted averages of training targets, with weights determined by kernel and training data.
- GP naturally handles noise and provides interpolation in the noise-free case, regression with uncertainty in the noisy case.